Designing for Reflective Creativity in Human-AI Interaction

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1. Introduction and Research Interest

I am applying for this PhD because I want to understand how we can design Generative AI to support slow and reflective creative work and how an AI tool can make space for thinking instead of pushing the user toward fast output. My interest in this topic comes from living in two worlds. Professionally, I work as a Machine Learning Engineer at Snappfood, where I build machine learning systems designed for speed and efficiency at scale. Personally, I care about mindful activities and art. I play music and practice coffee sensory tasting. In these practices, time is not an enemy. Slowness, care, and even a little effort are part of the meaning.

A common story today is that anyone can make art in seconds. A user types a prompt and gets an image at once. This is impressive, but it pushes creative practice toward fast production and output. It treats the creative process as a delay to be removed (Kemper and Jankowski, 2024). When the result is instant, the user can feel like a consumer, not an author. There is little room to think, to change direction, or to reflect. Some researchers even ask whether writing a prompt is really "making art" at all (McCormack et al., 2023).

I want to explore a different idea. Speed is not always the right goal for creative work. A creative tool can be designed to slow down, to leave room for reflection, and to keep the user in an active, thinking role. Research suggests that creative practice often benefits from qualities such as friction, indirection, and ambiguity (McNamara et al., 2025), and from slower and more careful ways of working. My research will study these ideas and ask how they can be turned into clear design principles for AI creativity support tools. I believe this can give users a stronger sense of ownership and agency over their creative work.

2. Research Background and Literature Review

The Efficiency Trap in Generative AI

For years, technology companies have designed their tools around seamlessness and immediacy, the goal is to remove friction so that every task feels fast and effortless. This works through **obfuscation**, the smooth surface of a tool hides the work, the data, and the costs behind it, leaving the user with only the fast result (Kemper and Jankowski, 2024). Generative AI inherits this philosophy and treats speed as a neutral, universal good. But for creative practice this framing has a cost, when a tool is built only for fast output, the creative process itself becomes a barrier to be removed, and attention shifts away from intention, exploration, and meaning.

Friction, Indirection, and Ambiguity as Resources

Not every difficulty is a problem. In design research, friction, indirection, and ambiguity are often treated as **resources**. Friction can slow the user down at the right moment and open space for reflection; indirection reaches a result through an indirect path rather than direct specification; and ambiguity leaves room for interpretation, so the user must think and decide for themselves. Gaver et al. (2003) showed that ambiguity can be a resource for design because it invites people to make their own meaning, and McNamara et al. (2025) show how interfaces built on metaphor and indirect control can open new ways of working.

Feminist AI and Small Data

Feminist perspectives on AI point out that large training datasets carry the biases of the society and the internet that produced them, so generative models are never neutral, they can repeat stereotypes and give even more weight to already-dominant styles, pushing quieter perspectives to the edge (Browne et al., 2023). The *small data* approach is one practical response. Bryan-Kinns et al. (2025) show that small, carefully chosen datasets can support more ethical and responsible creative AI, and that a bounded, curated dataset also makes the design space more intentional rather than an endless instant generator. This project takes a complementary route, it works with text-to-SVG tools and operationalizes reflective design through the structure of the creative process rather than through dataset curation. I acknowledge that this choice leaves a residual tension with the bias-amplification critique above, and I plan to revisit it in the discussion of the thesis.

Slow Technology

Slow technology is a design philosophy that values reflection over efficiency. Hallnäs and Redström (2001) frame design as a tool for thinking rather than for completing tasks quickly, challenging the assumption that faster is always better. Odom et al. (2014) extend this through **anticipation** and **re-visitation**, finding that when interactions unfold over time, users develop a deeper emotional attachment to the experience, waiting can be meaningful, not just frustrating.

The Gap

Together, these ideas point in the same direction, a slower, more careful, and more transparent way of working with AI is both possible and valuable for creative practice. Yet while these ideas are well discussed in HCI and design research, they have rarely been turned into concrete, working AI tools. Most research on generative AI still measures the output, how novel, realistic, or creative an image looks, and pays far less attention to the process. This project addresses that gap.

3. Research Questions

This PhD will investigate three core questions:

1. **Design principles.** How might a slower pace of working with AI for creative purposes be translated into design principles for creativity support tools?
2. **Productive vs. unproductive friction.** When users work with a slower AI tool, which forms of friction, indirection, and ambiguity genuinely support the creative process, and which only frustrate them?
3. **Agency and ownership.** How does interacting with a slower, more reflective AI tool affect users' sense of agency and creative ownership, compared with fast generation tools?

4. Research Methods

I will use a **Research-through-Design (RtD)** methodology (Zimmerman et al., 2007). This means I will build a working prototype to test my theoretical framework. The prototype itself becomes a research tool, generating insights through the process of design and user interaction.

4.1 Case Study: Slow Ink

My case study is called **Slow Ink**, a generative system for tattoo design that evolves over 7 days. I chose tattoos because they are permanent. This permanence naturally demands a slow, reflective design process. Slow Ink is a **research probe** for studying reflective creativity and agency in human-AI collaboration.

The role of journaling. Central to Slow Ink is a short daily journal. Each day, the participant writes a few sentences reflecting on their thoughts, mood, or what happened that day. The journal plays two roles at once. As a *reflective practice*, it slows the user down and invites them to sit with their design over the week. As a *system input*, it is the signal that guides each day's change, the system interprets the entry and uses it to evolve the open areas of the design, while the core stays fixed. In this way the user's own reflection becomes visible in the artwork, which is meant to support a sense of authorship rather than passive consumption.

How Slow Ink Works

Figure 1 gives an overview of the seven-day process.

- **Day 1:** The user provides a starting point in one of two ways: by writing a description of the tattoo they want, or by uploading an image of their own design which then we convert to editable SVG. The system produces a base design in vector form. The user then designates which parts are **anchor regions** and which surrounding or interior zones are **open regions** that the system may evolve, such as decorative patterns.
- **Days 2–7:** Each day, the system makes changes within the open regions only. The anchor regions stay fixed. The changes are guided mainly by the user's daily journaling.
- **Day 7:** The user reviews the whole seven-day journey. They can keep, revert, or further adjust any change.

Note: A key goal of the study is also to ask participants for their own ideas about what could shape the design, and to use that feedback when refining the system.

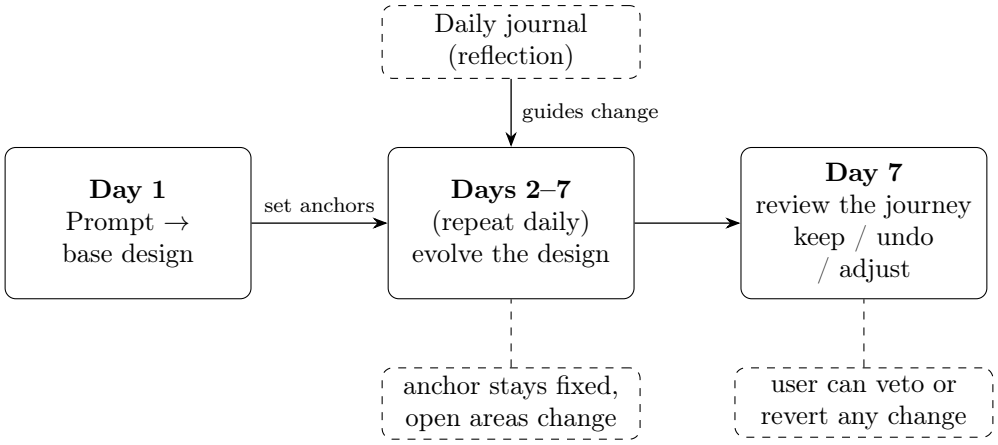


Figure 1: Overview of the seven-day Slow Ink process. A daily journal guides incremental changes to the open areas of the design while the anchor (core structure) stays fixed. The user keeps veto and revert control throughout.

Technical Implementation

The system works entirely in vector form (SVG) across the full 7-day cycle (Figure 2), which lets the core of the design be preserved exactly rather than approximately.

Day 1 (initialisation). A text prompt is turned into an SVG using LLM/VLM-based text-to-SVG methods such as OmniSVG (Yang et al., 2025) or SVGen (Wang et al., 2025), which output editable, layered path code rather than the over-parameterised, pixel-like shapes typical of diffusion-optimisation approaches (Jain et al., 2023); alternatively, an uploaded line drawing is converted into an SVG by vectorization, a route OmniSVG also supports directly through its image-to-SVG mode, so that the resulting paths are the participant's own drawing rather than a re-interpretation of it. Both routes are deliberately scoped to line-based, geometric tattoo styles rather than photorealistic or heavily shaded imagery; for these styles, vector representation is the natural fit, and this scoping also defines the class of tattoos the study addresses.

Anchor and open regions. The participant designates which parts of the Day-1 design are *anchor regions* (the frozen core) and which zones are *open regions* that the system may populate and evolve. Making this split an explicit user choice, both removes a hard segmentation problem and directly serves user agency, the participant decides what is fixed from the very first day. Recent text-to-vector work that learns ordered, layered shapes with standalone semantic meaning and exposes inference-time control over the rendered result (Polaczek et al., 2025) suggests this kind of shape-level decomposition is feasible to support directly in the representation rather than only as a post-processing step.

Days 2–7 (evolution). Each day, an LLM-based SVG editor modifies or adds path elements within the open regions only, conditioned on a lightweight signal derived from that day's journal entry. Specifying and validating the mapping from journal entry to concrete path edits is itself part of the planned technical work; in keeping with the transparency principle below, the interpreted signal is shown to the participant before it is applied, so they can accept or adjust it.

Local Processing: The system will utilize a local server or high-performance workstation physically located in the lab to ensure privacy, rather than using cloud APIs. Participants will interface with the system via a lightweight client on their laptop.

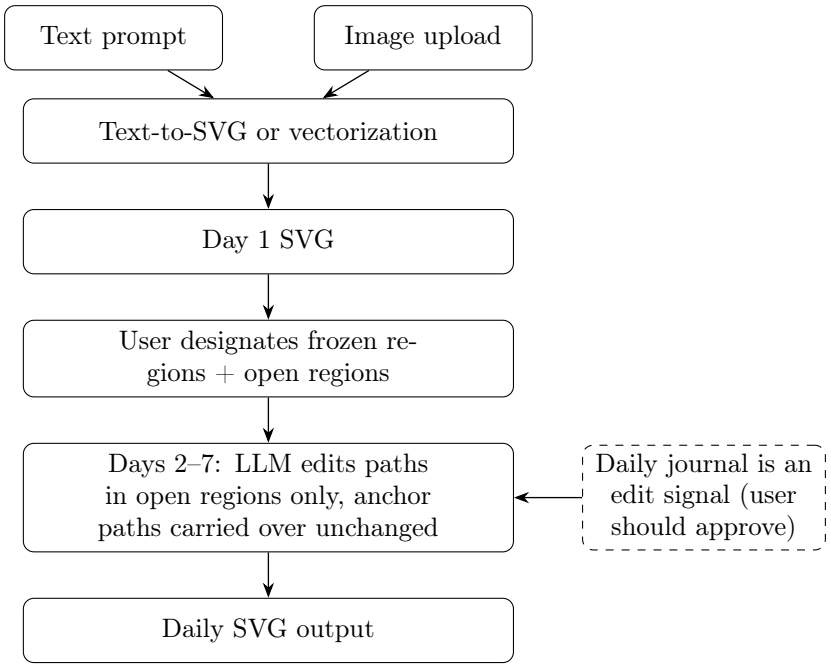


Figure 2: Technical pipeline (vector-native). The Day-1 design is produced as an SVG, either from a text prompt or by vectorizing an uploaded line drawing. The participant marks anchor and open regions; on Days 2–7 an LLM edits only the path elements in the open regions, conditioned on a signal derived from that day's journal, while anchor paths are carried over unchanged.

4.2 User Agency and Transparency

A core design principle of Slow Ink is **user control**. Users always know *why* the design changed through a transparent modification log. They retain **veto power** over any design change they do not like, and can selectively revert modifications from any point in the 7-day journey. This ensures that the AI acts as a collaborative partner rather than an autonomous agent, preserving user agency throughout the creative process.

4.3 Ethical Considerations and Data Protection

This study involves personal journal entries written by participants. I am committed to clear safeguards for participant privacy:

- **Informed Consent:** Before the study begins, participants receive a clear explanation of what data is collected and how it is used. Participation is voluntary, and they may withdraw at any time without giving a reason.
- **Data Minimization:** Journal entries and any derived labels are stored only for the duration of the study and deleted within 30 days of completion.
- **Ethics Approval:** Full ethical approval will be obtained from the the host university's research ethics committee before any data collection begins.

4.4 Evaluation Strategy

I will conduct a **comparative user study** with 15–20 participants. This sample size suits a mixed-methods study, it gives enough data to compare the two groups, while still allowing deep qualitative analysis (Creswell and Clark, 2017). Given the sample size, the quantitative measures below are treated as exploratory and serve to triangulate the qualitative findings, which carry the main analytical weight.

Participants: Adults who are interested in tattoos or body art.

Conditions: Both conditions run over the same 7-day window so that the contrast isolates the design of the system (state persistence, structural anchoring, controlled change) rather than the amount of time spent.

- *Control Group:* Uses a standard fast text-to-image tool over the 7 days. Within each daily session, participants can prompt and refine freely, but no state is carried over between days, so each day starts fresh. This represents a realistic fast workflow rather than a deliberately weak baseline.
- *Experimental Group:* Uses Slow Ink over the same 7 days.

Both groups keep daily journals, so that the contrast between conditions reflects the system design and not the presence or absence of a reflective writing prompt.

Output: At the end of the study, participants in both groups receive a **temporary tattoo sticker** of their final design. They can wear it for one to two weeks. This gives an embodied way to evaluate the result, without permanent commitment, and a short follow-up reflection captures how the design felt to live with.

Measures: The CSI and SoAS are administered at the end of the study.

- **Creativity Support Index (CSI)** (Cherry and Latulipe, 2014): a validated scale for how well a tool supports creative work.
- **Sense of Agency Scale (SoAS)** (Tapal et al., 2017): a validated scale for feelings of ownership and control.
- **Diary Studies:** participants keep daily journals reflecting on their experience.
- **Semi-Structured Interviews:** a 30-minute interview with each participant after the study. Interviews will also explore whether the slower process changed how participants thought about the *meaning* of their design, and whether any sense of ownership came from the slow pace, from the control they had, or from seeing their journal reflected in the design.

5. Research Timetable

Year 1 (Monthly Breakdown)

Month	Activity
1–3	Literature review: friction and reflective design; feminist AI; small data; slow technology
4–6	Develop the theoretical framework of design principles for reflective creativity support tools
7–10	Technical exploration: build the first prototype;
11–12	Pilot study with about 5 users; gather early feedback

Year 2 (Quarterly Breakdown)

Quarter	Activity
Q1	Refine the prototype based on pilot feedback
Q2	Recruit 15–20 participants; begin the main user study
Q3	Complete the main study; transcribe and code interviews
Q4	Analyze the quantitative data (CSI, SoAS)

Year 3 (Quarterly Breakdown)

Quarter	Activity
Q1	Synthesize findings; develop design principles for reflective creativity support tools
Q2	Write and submit a paper to a venue such as CHI or DIS
Q3	Write thesis chapters 1–3 (Introduction, Literature, Framework)
Q4	Write thesis chapters 4–5 (Case Study, Evaluation)

Year 4 (First Half)

Quarter	Activity
Q1	Write the final chapter (Discussion and Conclusion)
Q2	Revise the full thesis and submit for examination

6. Motivations and Career Aspirations

I am applying for this PhD because I believe the value of AI should not be measured by speed alone. There should be room for reflection, care, and creativity. My main aim is to support everyday users, people who are not professional artists (myself included) so that they feel a real sense of ownership over their creative choices.

My career goal is to become a researcher who builds AI systems that respect human creative processes. After completing this PhD, I hope to continue this work in an academic or industry research lab.

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